

OFFICE OF THE MEDICAL EXAMINER

AUTOPSY REPORT

Case Number:	ME-2024-001-ALK
Decedent Name:	CONFIDENTIAL
Age/Sex/Race:	42-year-old White Male
Date of Death:	November 13, 2024 (estimated)
Date of Autopsy:	November 16, 2024, 09:00 hours
Place of Autopsy:	County Medical Examiner Facility
Prosecutor:	Dr. Sarah Thompson, MD
Cause of Death:	Acute Ethanol Intoxication
Manner of Death:	ACCIDENT

CLINICAL SUMMARY

This 42-year-old male was discovered deceased in his apartment on November 15, 2024, by a neighbor who had not seen him for three days. The decedent was found lying prone on the living room floor surrounded by numerous empty alcohol containers. There were no signs of forced entry, struggle, or third-party involvement at the scene. The body demonstrated decomposition changes consistent with approximately 48-72 hours postmortem interval. Emergency services pronounced death at the scene.

EXTERNAL EXAMINATION

General Description

The body is that of a well-developed, adequately nourished adult male appearing consistent with the stated age of 42 years. The body measures 177 centimeters (5 feet 10 inches) in length and weighs 82 kilograms (181 pounds). The body is unclothed for examination purposes. Rigor mortis has resolved completely in all muscle groups. Livor mortis is present in posterior dependent areas and is fixed, consistent with the reported body position. Moderate decomposition changes are present

including skin slippage, marbling, and bloating consistent with the estimated postmortem interval of 48-72 hours.

Head and Face

The head is normocephalic and atraumatic. The scalp is intact without lacerations, contusions, or abrasions. Hair is brown, of medium length, with male pattern balding. The face shows no acute traumatic injuries. Perioral discoloration is noted with evidence of purge fluid around the mouth and nostrils, consistent with decomposition and terminal aspiration. Eyes demonstrate clouding of the corneas and flattening of the globes consistent with decomposition changes. Sclera are not icteric. Conjunctivae demonstrate petechial hemorrhages bilaterally. The nose is unremarkable without fracture or hemorrhage. The oral cavity contains partial upper and lower dentures. Natural teeth present show moderate dental decay but no acute trauma. The tongue is unremarkable.

Neck

The neck is symmetrical and shows no external evidence of trauma. There are no ligature marks, abrasions, contusions, or petechial hemorrhages visible on the skin surface. The trachea is palpably midline. Thyroid cartilage is intact without fracture. No masses or lymphadenopathy are palpable.

Trunk and Extremities

The chest is symmetrical without evidence of trauma. Abdomen is mildly protuberant with moderate bloating from decomposition gas formation. The skin demonstrates vascular prominence (spider angiomas) over the anterior chest and shoulders, and palmar erythema is noted on both hands, findings consistent with chronic alcohol use. Examination of both upper extremities reveals no defensive wounds, fractures, or acute traumatic injuries. No injection sites or track marks are identified. Fingernails are short and clean without tissue or debris beneath them. Both lower extremities are unremarkable without acute trauma. Minor well-healed scars are present on both knees consistent with remote injuries. External genitalia are normal adult male without acute pathology.

Evidence of Medical Intervention

No evidence of recent medical intervention or resuscitation attempts is present. No IV sites, ECG electrode marks, or defibrillation burns are identified.

INTERNAL EXAMINATION

Body Cavities

The body cavities are entered through a standard Y-shaped incision. The subcutaneous tissue shows moderate adiposity. The musculature is well-developed and red-brown in color. The pleural cavities contain approximately 50 mL of clear serous fluid bilaterally. No adhesions are present. The pericardial sac contains approximately 30 mL of clear serous fluid. The peritoneal cavity contains approximately 100 mL of turbid fluid with early decomposition changes. No blood or other abnormal fluid collections are identified. The organs are in normal anatomic position without evidence of surgical alteration.

Cardiovascular System

The heart weighs 380 grams (normal range 270-360 grams). The epicardial surface is smooth and glistening without adhesions. The coronary arteries arise normally from the aortic root and follow normal distribution patterns. Serial sectioning of the coronary arteries reveals mild atherosclerotic changes with luminal narrowing estimated at less than 30% in all vessels. No acute thrombosis, plaque rupture, or significant stenosis is identified. The myocardium is uniformly red-brown without areas of pallor, hemorrhage, or scarring to suggest acute or old infarction. Ventricular walls measure: left ventricle 1.3 cm (normal 0.9-1.4 cm), right ventricle 0.3 cm (normal 0.2-0.4 cm). All four cardiac chambers are of normal size and configuration. The endocardium is smooth and glistening. All cardiac valves (aortic, mitral, tricuspid, pulmonic) are thin, pliable, and competent without vegetations, calcifications, or abnormalities. The aorta demonstrates minimal atherosclerotic changes without aneurysm or dissection. The superior and inferior vena cavae are patent and unremarkable.

Respiratory System

The larynx and trachea are patent and lined with smooth, pink-tan mucosa. No obstruction, edema, or hemorrhage is present. The hyoid bone and thyroid cartilage are intact without fracture. Bronchi are patent and contain aspirated gastric contents mixed with mucus. The right lung weighs 620 grams (normal 450 grams) and the left lung weighs 620 grams (normal 400 grams). Both lungs are moderately congested and edematous with pink-tan parenchyma. On sectioning, frothy pink fluid exudes from the cut surfaces, consistent with pulmonary edema. Scattered areas of red-purple discoloration represent hypostatic congestion. Aspirated gastric contents are visible within bronchi and bronchioles. No masses, nodules, consolidations, or emboli are identified. The pleural surfaces are smooth without adhesions or tumor.

Hepatobiliary System

The liver weighs 2,180 grams, significantly enlarged (normal 1,400-1,800 grams). The capsular surface is smooth but the organ has a yellow-brown appearance consistent with fatty infiltration. The consistency is soft and friable. On sectioning, the cut surfaces demonstrate diffuse yellow discoloration with loss of normal lobular architecture, classic findings of hepatic steatosis (fatty liver disease). The degree of fatty change is estimated at greater than 70% of hepatocytes grossly. No masses, cysts, abscesses, or focal lesions are identified. The gallbladder contains approximately 20 mL of dark green bile and no stones. The mucosa is velvety and unremarkable. The extrahepatic biliary tree is patent without obstruction.

Gastrointestinal System

The esophagus is patent and lined with gray-white mucosa without varices, ulceration, or perforation. The stomach is partially distended and contains approximately 300 mL of liquid with a strong alcoholic odor mixed with partially digested food material. The gastric mucosa demonstrates diffuse erythema (redness) and multiple small superficial erosions consistent with acute alcoholic gastritis. No ulcers, masses, or perforations are present. The small intestine is unremarkable with normal mucosal folds and no obstruction, perforation, or masses. The appendix is identified and unremarkable. The large intestine contains formed brown stool without blood. The mucosa is unremarkable without masses, polyps, or inflammation. The pancreas weighs 110 grams (normal 70-100 grams) and is tan-yellow with a soft, lobular consistency. No masses, cysts, or hemorrhage are identified. Focal areas of fat necrosis suggest chronic pancreatitis related to alcohol use.

Genitourinary System

The right kidney weighs 160 grams and the left kidney weighs 165 grams (combined normal 240-340 grams). Both kidneys have smooth capsular surfaces that strip easily. The cortical surfaces are red-brown with normal corticomedullary junction. No masses, cysts, stones, or hydronephrosis are present. The renal pelves and ureters are unremarkable. The urinary bladder contains approximately 150 mL of cloudy urine. The mucosa is unremarkable. The prostate gland is normal in size and consistency without masses or nodules. The testes are descended bilaterally and unremarkable.

Endocrine System

The thyroid gland is normal in size, shape, and consistency without masses, nodules, or cysts. The adrenal glands are normal in size with distinct cortex and medulla without hemorrhage or masses. The pituitary gland is unremarkable.

Hemolymphatic System

The spleen weighs 180 grams (normal 100-250 grams). The capsule is intact and smooth. The parenchyma is red-purple and firm with normal trabecular pattern. No masses or infarcts are identified. Lymph nodes throughout the body are not enlarged and appear normal.

Central Nervous System

The scalp is reflected in standard fashion revealing no subgaleal hemorrhages. The skull is intact without fractures, depressions, or defects. The skull is opened with an oscillating saw. The dura mater is intact, smooth, and glistening without adhesions or tears. No epidural or subdural hemorrhages are present. The dura is reflected to reveal the brain. The leptomeninges are clear and delicate without exudate, hemorrhage, or adhesions. The brain weighs 1,420 grams (normal 1,300-1,400 grams). The cerebral hemispheres are symmetrical without mass effect, shift, or herniation. The gyri are normal in size and configuration. The sulci are not widened. No areas of hemorrhage, contusion, laceration, or mass lesions are visible on the external surface. The blood vessels at the base of the brain (Circle of Willis) are intact and patent without aneurysms, atherosclerosis, or thrombosis. The cranial nerves are identified and appear normal. The brain is sectioned in the coronal plane at 1 cm intervals. The sectioned surfaces reveal normal gray-white matter differentiation. The cerebral cortex, white matter, basal ganglia, and thalami appear normal without hemorrhage, infarction, or mass lesions. The ventricular system is of normal size and configuration. The cerebellum and brainstem are unremarkable without herniation. The spinal cord is not examined.

Musculoskeletal System

The ribs are examined by palpation and demonstrate no fractures. The spine is intact without fractures or subluxations. Long bones of the extremities are intact without fractures. Joint spaces are unremarkable. Skeletal muscle is well-developed with normal red-brown color.

MICROSCOPIC EXAMINATION

Representative sections of organs are submitted for microscopic examination and processed in routine fashion with hematoxylin and eosin staining.

Liver: Microscopic examination confirms extensive macrovesicular steatosis affecting greater than 70% of hepatocytes. Large fat vacuoles displace nuclei peripherally. Focal areas show inflammatory infiltrate consisting of neutrophils and lymphocytes, consistent with early alcoholic hepatitis. Mild periportal fibrosis is present. No cirrhosis is identified, though the changes indicate progression in that direction with continued alcohol exposure.

Lungs: Sections demonstrate moderate pulmonary edema with proteinaceous fluid within alveolar spaces. Aspirated gastric contents are visible within bronchioles and alveolar spaces, accompanied by acute inflammatory response. No pneumonia, fibrosis, or neoplasm is identified. Pulmonary vessels show no emboli.

Heart: Myocardial fibers are uniform in size without hypertrophy or disarray. No acute injury, necrosis, fibrosis, or inflammation is present. Coronary artery sections show minimal atherosclerotic changes without significant stenosis.

Brain: Sections from cerebral cortex, basal ganglia, and cerebellum show normal neuronal architecture without acute injury. Mild cerebral edema is present. No inflammation, necrosis, infarction, or pathologic changes are identified.

Kidneys: Glomeruli, tubules, interstitium, and vessels appear within normal limits. No acute tubular necrosis, glomerulonephritis, or significant pathology is present.

Pancreas: Focal areas of fat necrosis with chronic inflammation consistent with chronic pancreatitis related to alcohol use. No acute pancreatitis is present.

OPINION

Based on the autopsy findings, toxicological results (blood alcohol concentration 0.42%), scene investigation, and medical history review, it is my opinion that the cause of death is **acute ethanol intoxication**. The manner of death is **accident**.

The decedent died as a result of the toxic effects of extremely elevated blood alcohol concentration. The mechanism of death involved central nervous system depression leading to respiratory depression, loss of protective airway reflexes, aspiration of gastric contents, and cardiorespiratory failure. The autopsy findings of pulmonary edema, aspiration pneumonitis, and severe alcoholic fatty liver disease support this conclusion. The absence of significant trauma, natural disease processes sufficient to cause death, or evidence of foul play supports the classification as accidental death.

Sarah Thompson, MD

Chief Medical Examiner

Board Certified in Anatomic and Forensic Pathology

Date: November 18, 2024